

Question	Marking Guidance	Mark	Comments
09.1	1. Chromosomes/centromeres cannot attach (to spindle) OR Chromosomes cannot line up (on spindle); 2. (So, no) metaphase; OR 3. <u>Chromatids</u> cannot separate (on spindle); 4. (So, no) anaphase;	2	Mark in pairs as (1 and 2 OR 3 and 4) 3. Accept description of 'cannot separate' e.g cannot move to poles 3. Ignore 'split'
09.2	1. Cancer cells divide more/uncontrollably/rapidly OR Healthy cells divide less/slowly;	1	
09.3	1. (ABZ) increases/maintains Cyclin B; 2. (So) <u>mitosis</u> (starts but) does not end (no tumour growth); 3. (ABZ) lowers ratio of Bcl-2 to Bax; 4. (So) apoptosis occurs/cells die (no tumour growth);	4	Ignore references to spindle fibres 2. Ignore mitosis will continue 2. Ignore mitosis stops without qualification. 3. Accept (ABZ) decreases Bcl-2 and increases Bax
09.4	In support of suggestion 1. Stops mitosis (at metaphase/anaphase)/cell division (so no tumour growth) OR Promotes apoptosis/programmed cell death (so tumour destroyed); Against suggestion 2. Healthy cells (are) damaged/affected OR Causes side effects;	3 max	2 max for points "Against suggestion"

	3. Results from laboratory tests/tests on (isolated) cells OR No clinical trials/tests on patients/people; 4. Dosage unknown; 5. Effectiveness unknown;		
TOTAL		10	